

Identifying twitter profiles that are mistakenly recorded as human/non-human profiles

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Abstract

This project aims to identify and correct profiles on Twitter that are mistakenly recorded as either human or non-human (brand) using machine learning and big data analytics techniques. By following the Big Data Analytics Lifecycle, the dataset undergoes preprocessing, feature transformation, and modeling to accurately classify users based on their behavior, profile characteristics, and engagement metrics. The analysis includes the application of various machine learning algorithms such as Logistic Regression, Decision Trees, K-Nearest Neighbors, and Random Forests, supported by Python-based tools and libraries. Core models like classification, clustering, and text processing methods are employed to analyze and classify the profiles effectively. Visualization techniques are used to explore the data and evaluate model performance. The results of the study provide insights into patterns that distinguish human profiles from non-human (brand) profiles, addressing potential misclassifications and improving the accuracy of profile identification on Twitter. The findings contribute to understanding misinformation in social networks by correcting incorrect profile classifications.

1 Introduction

In today's digital landscape, social media platforms like Twitter have become integral to communication, marketing, and information dissemination. However, distinguishing between human users and non-human profiles, such as brand or organization accounts, is crucial for many applications, including targeted marketing, social behavior analysis, and combating misinformation. This project focuses on the classification of Twitter user profiles into two categories: human (male or female) and non-human (brand), using ma-

chine learning techniques.

By leveraging the Big Data Analytics Lifecycle, this project begins with data exploration and preprocessing, ensuring the dataset is cleaned and transformed for analysis. Feature engineering, including the handling of text data, categorical transformations, and numerical features, plays a key role in preparing the dataset for classification tasks. The objective is to accurately classify user profiles based on various features such as tweet content, engagement metrics (likes, retweets), and user behavior patterns.

Machine learning models, including Logistic Re-

gression, Decision Trees, Random Forests, and K-Nearest Neighbors, are applied to distinguish between human and non-human profiles. Furthermore, an analysis is conducted to identify and correct misclassifications in the dataset, improving the accuracy of the models. By following this structured approach, the project aims to provide insights into the characteristics of Twitter profiles and ensure accurate classification.

2 Data Discovery

In the discovery phase of this project, the objective is to gain a comprehensive understanding of the dataset and to define the goals for the analysis. The dataset consists of 20,050 entries. Each representing a unique Twitter user profile with various attributes such as gender, tweet content, profile metadata, and other user-related features. These attributes span multiple data types, including boolean, categorical, numerical, and textual data.

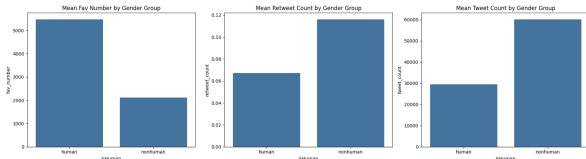


Figure 1:

The primary focus of this project is to identify Twitter profiles that are misclassified as human or non-human. This classification is essential, as errors can affect various downstream applications, including content moderation, user behavior analysis, and marketing strategies. The dataset provides a diverse set of features to facilitate this analysis:

Demographic Information: The dataset includes columns such as *gender* and *gender:confidence*, which provide insights into the predicted gender of users. There are approximately 100 missing values in the *gender* column, which will need to be addressed during the data preparation phase.

Profile Metadata: The *profile_yr*, *fav_number*, *retweet_count*, and *tweet_count* columns contain user profile information that may be indicative of user behavior. These features could play a crucial role in differentiating between human and non-human profiles.

Textual Data: The text and description columns contain user-generated content, including tweets and profile descriptions. Analyzing this textual data through methods such as Term Frequency-Inverse Document Frequency (TF-IDF) and natural language processing (NLP) will be key to extracting meaningful patterns.

Customization Preferences: The dataset also includes customization preferences like *link_color* and *sidebar_color*, which may be useful for clustering or as additional features in predictive modeling tasks.

Geolocation and Timezone Information: Features such as *tweet_location* and *user_timezone* offer geographical context for user activity. However, these columns contain a significant amount of missing data, which will need to be addressed.

The discovery phase allows for the formulation of a hypothesis that both behavioral and textual features will significantly contribute to identifying incorrectly classified profiles. This project will involve employing a combination of data cleaning, feature engineering, and machine learning techniques, with a focus on classification algorithms.

Following the discovery stage, the data will be prepared by addressing missing values, handling outliers, and performing feature engineering. Model planning and building will involve evaluating various machine learning algorithms such as decision trees and logistic regression to determine their effectiveness in solving the classification problem. Finally, the results will be analyzed and visualized to assess model performance and inform potential improvements.

3 Data Preparation

In the data preparation stage, the dataset undergoes several transformations and cleaning processes to ensure it is suitable for modeling. The main objective is to refine the data by addressing missing values, removing irrelevant features, transforming text and categorical data, and converting date-time values for use in machine learning models. This stage focuses on ensuring that the data is clean, consistent, and ready for model building.

3.1 Feature Dropping

To begin, several features are dropped from the dataset as they do not provide any meaningful information for the analysis. These features include:

- **Identifiers:** Features like *unit_id* and *tweet_id*, which are user and tweet IDs, are dropped because they do not contribute to the classification task.
- **Duplicate Information:** Features such as *unit_state*, *trusted_judgement*, *last_judgement*, *gender_gold*, and *profile_yn_gold* are removed as they contain redundant information already covered by the *golden* column.
- **URLs:** The *profileimage* column, which contains URLs to profile images, is also dropped because URLs are irrelevant to the classification task.
- **Constant Values:** The *tweet_created* column is removed as it contains a constant value for all entries, offering no variability for the analysis.
- **Names:** The *name* Contains no useful information for machine learning, possibly because names are not predictive of the target and might introduce noise or over-fitting.
- **Geo-location and Timezone Information:** The *tweet_location* Has too many missing values, making it unreliable for the model. Instead of imputing or filling in those values, it's more efficient to drop the feature altogether.

After dropping these features, the dataset retains 14 features and 1 label (target), the "*gender*" column. The data now consists of a mixture of text, numeric, categorical, boolean, and geographic data, as well as a date-time feature.

Since the "unknown"s and "blanks"s cannot be used as label, those entries will be dropped.

3.2 Handling Missing Values

The dataset contains missing values in several columns, primarily in text fields. These missing values are handled as follows:

- **Text Fields:** Missing values in columns such as *description*, *tweet_location*, and *user_timezone* are replaced with the string "not available", ensuring that null values do not cause issues during analysis.
- **Numeric Fields:** The *gender:confidence* column, which has some missing values, is filled with a neutral value of 0.5, indicating uncertainty in the confidence of gender prediction.

This step ensures that the dataset does not contain any null values that could disrupt the analysis or model training.

3.3 Text Feature Transformation

: The dataset includes several text-based columns: *description*, *name*, and *text*. These features undergo the following transformations:

- **Cleaning:** All text data is cleaned to remove unnecessary characters, punctuation, and symbols that could interfere with analysis.
- **Lemmatization & Stemming:** The text is further processed to reduce words to their base forms (lemmatization) and root forms (stemming) to enhance the consistency of the vocabulary.
- **TF-IDF Vectorization:** To convert text data into numerical form, Term Frequency-Inverse Document Frequency (TF-IDF) is applied to create sparse matrices representing the importance of words in each text feature. These numerical representations will be used in the modeling process.

The vocabulary was crated from only top 90% token in terms of importance denoted by TF-IDF values for both text features.?

3.4 Categorical Feature Transformation

Two categorical features, *link_color* and *sidebar_color*, are stored in hexadecimal format. These features undergo the following transformations:

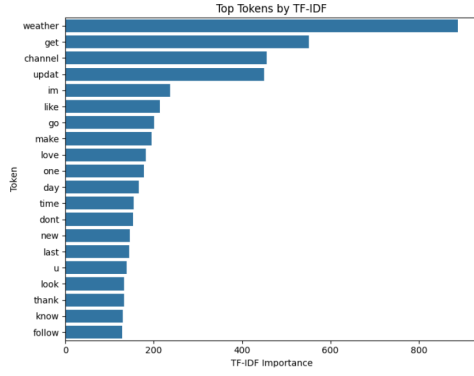


Figure 2: Most important words

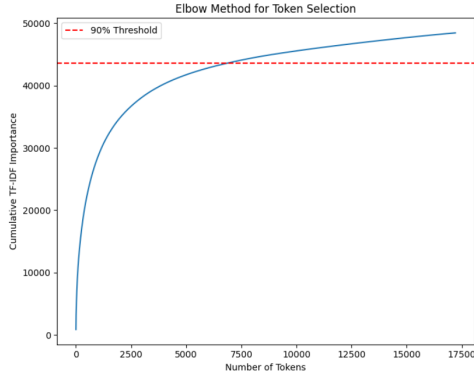


Figure 3: Cumulative TF-IDF elbow

- **Hexadecimal Cleaning:** The hex color values are cleaned to remove inconsistencies. This involves standardizing the values to ensure they are valid 6-character hex codes.
- **RGB Conversion:** After cleaning up the hex values, they are turned into RGB (Red, Green, Blue) numbers. This makes it easier for machine learning models to use the color data because the colors are now in numerical form. This way, the color info is preserved and turned into numbers for the models.

3.5 Date-Time Feature Transformation

The data contains one date-time feature, created, which records the account creation date. This feature is transformed as follows:

Account Age: The account creation date is converted into the account's age, measured in years, by calculating the difference between the current date and the account creation date. This transformation creates a meaningful numeric feature that reflects how long the account has existed.

3.6 Boolean Feature Transformation

Two boolean features, *golden* and *profile_yn*, are present in the dataset. The feature *golden* are transformed into binary values (0 and 1). Since *profile_yn:confodence* is probability associated to *profile_yn*, the boolean field is encoded using yes:1 and no:-1 and then multiplied to *profile_yn:confodence*. This way the probality information is kept which would not have been possible for "no" values had the field been encoded to 0 and 1.

3.7 Label Transformation

The objective of the project is to distinguish between human and non-human Twitter profiles. The dataset's gender column is used as the label, where entries labeled as "male" or "female" represent human users, and entries labeled as "brand" represent non-human users.

Handling Unusable Labels: Entries with missing, unknown, or blank values in the gender column are dropped, as these cannot be used as part of the classification task. **Class Consolidation:** The gender column is transformed into a binary label where human profiles (male, female) are coded as 1 and non-human profiles (brands) are coded as 0. With these transformations completed, the data is now ready for model building. The preparation process has refined the dataset into a form that is clean, free of missing values, and fully numeric, enabling efficient model training and accurate predictions.

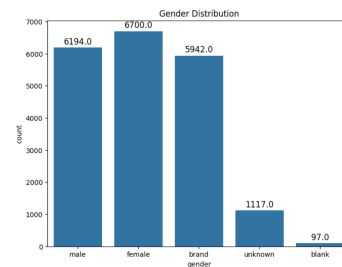


Figure 4: Before dropping the unknown and blank data

4 Model Planning and Building

The model planning and building stage is essential to select appropriate models, split the data into train-

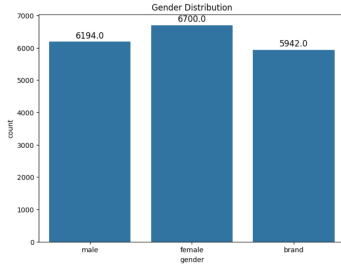


Figure 5: After dropping the unknown and blank data

ing and testing sets, and scale the features to ensure consistent input for the classification models. The objective of this project is to distinguish between human and non-human profiles based on the available features. This stage outlines how the dataset was prepared for machine learning models and how the data was split to maintain the distribution of human and non-human profiles.

4.1 Data Splitting and Stratification

The dataset was split into training and testing sets with a test size of 20%, ensuring that 80% of the data was used for training the model, while 20% was reserved for evaluating model performance. Given that 32% of the profiles in the dataset were labeled as non-human, stratified splitting was performed based on the label (isHuman). This ensured that the proportion of human and non-human profiles was consistent across both the training and testing sets.

Target Variable (y): The target variable was defined as $y = df['isHuman']$, representing whether a profile is human (1) or non-human (0). **Feature Variable (X):** The feature set was prepared by dropping the isHuman column, leaving the remaining features for use in model training.

4.1.1 Feature Scaling

The dataset contains a mix of numeric features, color values in RGB format, and text features. Scaling was performed to standardize the numeric features to ensure that all features had a similar range, improving model performance. **StandardScaler** was used to transform these features.

The numeric features were grouped as follows:

Color Features: The link_color and sidebar_color

columns were split into their respective RGB components (li_R, li_G, li_B, sb_R, sb_G, sb_B), which were then scaled.

Count Features: The fav_number, retweet_count, and tweet_count columns represent various counts related to user activity, which were scaled to normalize their distributions.

Full Dataset: The entire feature set, including text and numeric data, was scaled and prepared for model building.

4.1.2 Text and Dense Data Split

Since the dataset contains both text-based and numeric features, the data was divided into separate categories for testing the models:

- **Text Data (Sparse matrix):** Features such as the tweet and description columns were vectorized using TF-IDF and combined into a sparse matrix representation. This was used for text-specific models.
- **Dense Data (Non-Text):** Numeric features were scaled and combined separately to create dense feature sets, excluding text data.

The final input matrices for the models were constructed as follows:

1. **Full Data:** A combination of the scaled numeric features and TF-IDF vectorized text data.
2. **Text Data Only:** A dataset consisting solely of the TF-IDF matrix of tweet and description.
3. **Dense Data Only (Excluding Text):** A dataset of all scaled numeric features.
4. **Color Data Only:** A dataset containing only the scaled RGB values of link_color and sidebar_color.
5. **Count Data Only:** A dataset consisting only of the scaled count features (fav_number, retweet_count, tweet_count).

Each subset was split into training and testing sets using the train_test_split() function, maintaining the

same 80-20 split ratio with stratification on the target label.

Classification Models Several classification models were selected to predict whether a profile is human or non-human. These models were chosen for their performance on similar tasks and their ability to handle both sparse and dense data. The following models were employed:

Logistic Regression: A linear classifier often used as a baseline model for binary classification tasks.

K-Nearest Neighbors (KNN): A non-parametric model that classifies data points based on the majority label of their nearest neighbors. KNN is relatively slower than other models. Hence, only color data and count data is trained and tested on this model.

Decision Tree: A non-linear model that creates a tree-like structure to predict the class label based on feature splits.

Random Forest: An ensemble of decision trees, where multiple trees vote on the final classification, improving generalization.

Naive Bayes: A probabilistic model that assumes independence between features, often used for text classification. The text data split is trained and tested with MultinomialNB because the model is particularly well-suited for text data and TF-IDF sparse matrices. Results below

Multi-layer Perceptron (MLP): A neural network-based model that uses multiple layers of perceptrons to learn complex patterns in the data.

Support Vector Machine (SVM): A powerful classifier that attempts to maximize the margin between the classes in a high-dimensional space.

Each model was trained and tested using the various subsets of the data (full, text-only, dense-only, color-only, count-only) to determine which feature set and model combination yields the best classification results.

In conclusion, the model planning and building stage involved careful data splitting and feature scaling, followed by the application of multiple classification models. The next step is to evaluate each model's performance to identify the most effective approach for distinguishing between human and non-human profiles.

4.2 Experiment 1

Full Data trained and tested on all of the below models:

- Logistic Regression
- KNN
- Decision Tree
- Random Forest
- Support Vector Machine

4.3 Experiment 2

Since Logistic Regression is the fastest among all the models, all splits of data were trained and tested separately on Logistic Regression

4.4 Experiment 3

The text data split is trained and tested with MultinomialNB because the model is particularly well-suited for text data and TF-IDF sparse matrices.

4.5 Experiment 4

Only Color data were trained and tested with SVM model

4.6 Experiment 5

Only Counts data trained and tested with Decision Tree

5 Result & Discussion

5.1 Result: Experiment 1

Evaluation metrics from All models on Full Data.

	Accuracy	Precision	Recall	F1
LogReg	82.35	82.27	94.61	88.01
KNN	76.88	78.26	91.70	84.45
DT	78.16	82.52	86.39	84.41
RF	83.41	81.80	97.44	88.94
SVM	79.72	77.81	98.45	86.92

Table 1: Evaluation metrics on Full Data

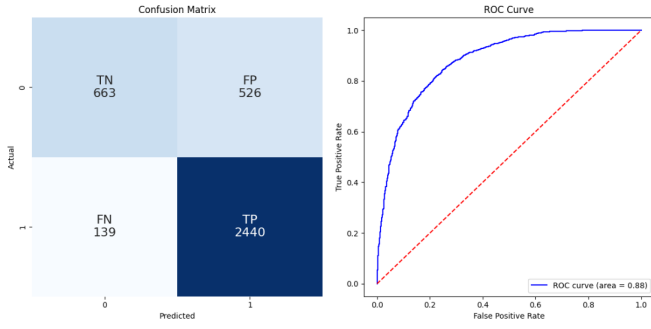


Figure 6: LogReg CM and ROC

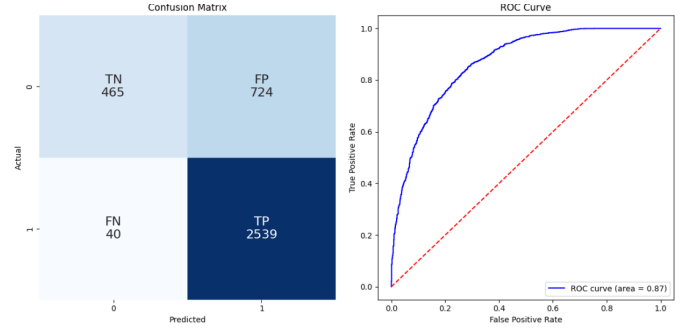


Figure 10: LogReg CM and ROC

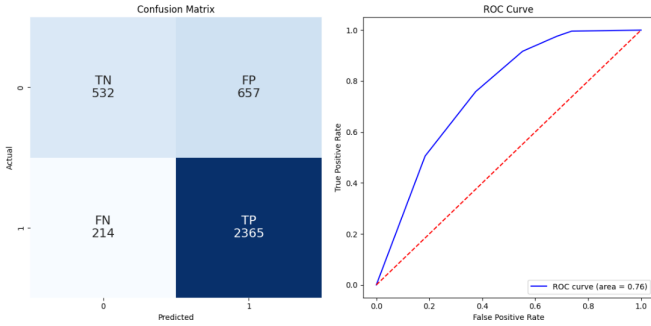


Figure 7: LogReg CM and ROC

5.2 Result: Experiment 2

Log Reg evaluation metrics on all splits of the dataset.

	Accuracy	Precision	Recall	F1
FullData	82.35	82.27	94.61	88.01
Text	83.15	82.12	96.36	88.67
Dense	68.95	69.23	98.33	81.26
Color	68.44	68.44	1.00	81.27
Counts	68.84	69.07	98.64	81.25

Table 2: LogReg metrics on different splits of the data

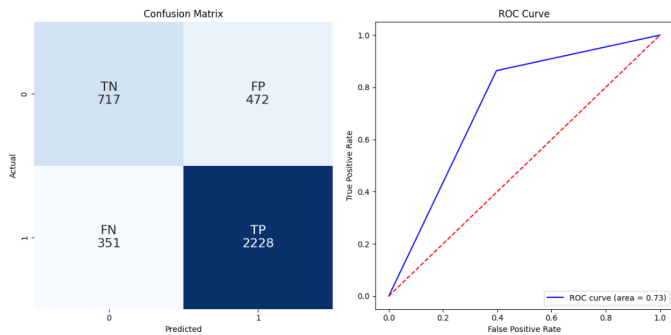


Figure 8: LogReg CM and ROC

5.3 Result: Experiment 3

MultinomialNB metrics trained and tested on text data only.

- Accuracy: 0.8248
- Precision: 0.8202
- Recall: 0.9531
- F1 Score: 0.8816

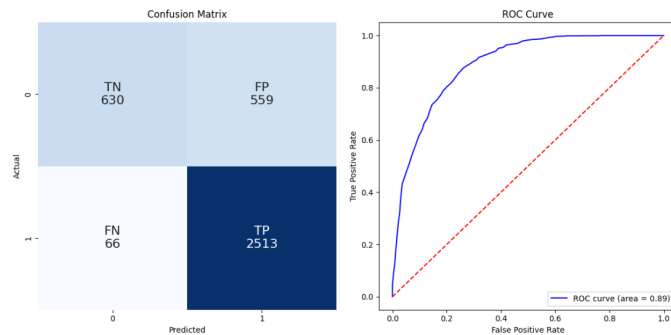


Figure 9: LogReg CM and ROC

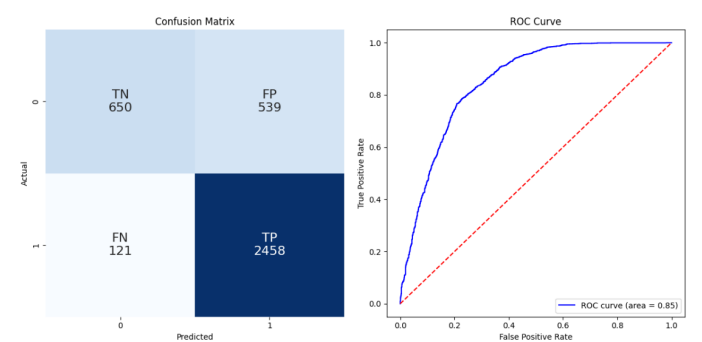


Figure 11: CM and ROC by MultinomialNB on text data

5.4 Result: Experiment 4

Only Color data trained and tested with SVM.

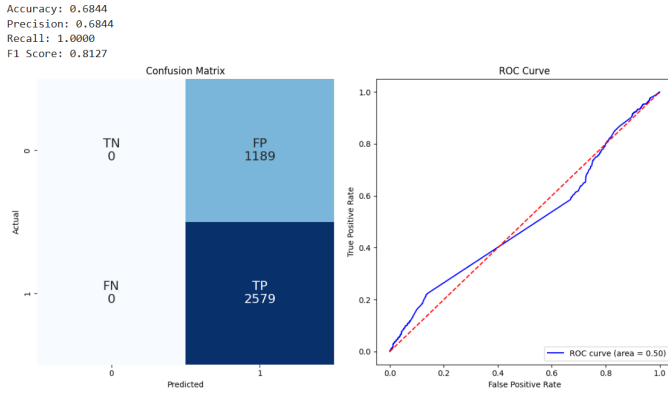


Figure 12: CM and ROC on color data only

5.5 Result: Experiment 5

Only count data trained and tested with KNN.

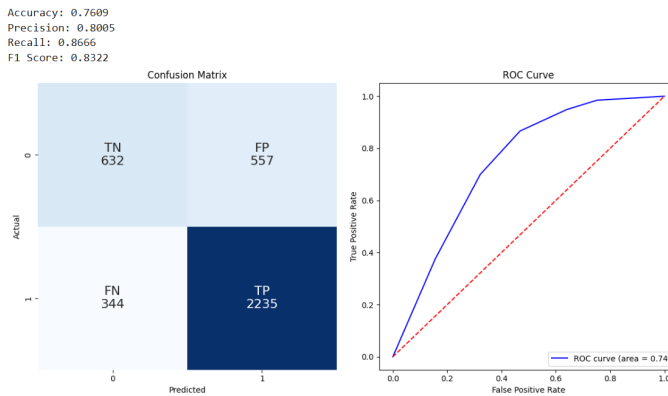


Figure 13: CM and ROC on count data only

6 Discussion

The experiments in this project helped to show how well different machine learning models can classify Twitter profiles as either human or non-human. By using various feature sets like text, numeric, color, and count data, the results highlighted which features and models worked best for this task.

Model Performance on Full Data When all models were tested on the full dataset (which included text and numeric features), Random Forest performed the best, with an accuracy of 83.41% and an F1 score of 88.94. This model worked well because it combines many decision trees, which helps to make better predictions. Logistic Regression also did well, with an accuracy of 82.35% and an F1 score of 88.01%, showing that it's a simple but effective model for this type of problem. Other models like SVM and K-Nearest

Neighbors (KNN) also did okay, but they didn't perform as well as Random Forest and Logistic Regression.

This suggests that ensemble models like Random Forest are great at handling more complex data, while linear models like Logistic Regression are still reliable and fast. Models like SVM and Decision Trees showed some potential, but they didn't beat Random Forest.

How Feature Subsets Affected Results In the second experiment, Logistic Regression was used on different types of data (text, numeric, color, and count). The text data gave the best results, with an accuracy of 83.15% and an F1 score of 88.67%. This was even better than using all the features together. This shows that the words in tweets and descriptions are really important for telling if a profile is human or not.

The numeric data, color data, and count data didn't do as well, with accuracies around 68-69%. This means that while they add some value, they aren't as good on their own as the text features. The color and count data didn't provide much helpful information about whether a user was human or a brand.

Multinomial Naive Bayes on Text Data In the third experiment, Multinomial Naive Bayes (MNB) was applied to the text data subset, achieving an accuracy of 82.48% and an F1 score of 88.16%. These results are comparable to Logistic Regression, indicating that Naive Bayes is well-suited for text-based classification tasks, particularly when using TF-IDF sparse matrices. MNB's assumption of feature independence works well in text classification, where the frequency of words (represented by TF-IDF values) can provide strong signals for classification.

Effectiveness of Individual Feature Types In experiments four and five, the SVM and Decision Tree models were trained and tested on color data and count data, respectively. The results indicated that these individual feature types provided relatively low accuracy compared to the text data and full data models.

7 Conclusion

Overall, the experiments showed that text features are the most important for figuring out if a Twitter profile

is human or non-human. The best-performing models, like Random Forest and Logistic Regression, were able to use these text features to make strong predictions. While numeric, color, and count data did contribute, they weren't as effective as the text data. Random Forest stood out as the best model, while Logistic Regression was a reliable option that performed well across all tests.

In summary, using text data with models like Logistic Regression and Random Forest gives the best results for classifying Twitter profiles. In the future, the models could be further fine-tuned, or more advanced text processing techniques could be tried to improve the results even more.

8 References

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